ModelIt!

Teacher's Guide

Can the Ocean Power Entire Cities?

Modeling Ocean Thermal Energy Conversion and Marine Renewable Power

Quick Reference

Grade Level	Grade 10L3
NGSS Standard	HS-PS3-2, HS-ESS3-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-PS3-2, HS-ESS3-1

Develop and use models to illustrate that energy at the macroscopic scale can be accounted for as a combination of energy associated with the motions of particles (objects) and energy associated with the relative positions of particles (objects). Construct an explanation based on evidence for how the availability of natural resources, occurrence of natural hazards, and changes in climate have influenced human activity.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop thermodynamic models illustrating energy conversion from macroscopic thermal energy to electrical energy, accounting for efficiency limits and energy losses at each stage.
Disciplinary Core Idea	**PS3.B Conservation of Energy and Energy Transfer / ESS3.A Natural Resources** — Students model how thermal energy stored in ocean temperature gradients can be converted to useful work within thermodynamic constraints, and evaluate ocean thermal energy as a natural resource.
Crosscutting Concept	**Energy and Matter** — Students trace energy flow from solar radiation absorbed by ocean surface water through the OTEC thermodynamic cycle to electrical output, quantifying conversion efficiency and losses at each stage.

CAST Testing Alignment

MULTIPLE CHOICE:

An OTEC plant operates with a surface water temperature of 27 degrees Celsius and deep water temperature of 5 degrees Celsius. The Carnot efficiency is approximately 7.3%. If the plant processes thermal energy equivalent to 500 MW, what is the maximum possible electrical output, and why is the actual output even lower?

CONSTRUCTED RESPONSE:

Using your model, explain why OTEC requires pumping millions of gallons of water per minute to generate power that a natural gas plant produces with a relatively small fuel supply. Connect your explanation to the Carnot efficiency limit and the small temperature differential.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The ocean is the largest solar collector on Earth. Tropical surface waters absorb enough solar energy to power human civilization hundreds of times over. Just below the surface, at depths of 800-1,000 meters, water temperatures hover near 4 degrees Celsius — essentially unchanged for centuries. This temperature difference between surface and deep water represents an enormous energy reservoir that Ocean Thermal Energy Conversion technology can tap — but thermodynamics imposes strict limits on how much energy can actually be extracted.

The OTEC Thermodynamic Cycle

OTEC uses a closed-cycle Rankine heat engine. Warm surface water (25-30 degrees Celsius) heats a working fluid (usually ammonia, which boils at -33 degrees Celsius) in an evaporator. The ammonia vapor expands through a turbine, generating mechanical energy that drives a generator. Cold deep water (4-5 degrees Celsius) then condenses the ammonia vapor back to liquid in a condenser. The liquid ammonia is pumped back to the evaporator, completing the cycle. The critical constraint is the Carnot efficiency: with only a 20 degree temperature difference, the theoretical maximum efficiency is about 6.7%.

The Pumping Challenge

Because efficiency is so low, OTEC must process enormous volumes of water. A 100 MW net output plant requires pumping approximately 400 cubic meters per second of warm water and 200 cubic meters per second of cold water — comparable to a medium-sized river. The cold water pipe alone extends 1,000+ meters to the deep ocean and may be 8-10 meters in diameter. Pumping this much water consumes 30-40% of the gross power generated, which is why net efficiency drops from the theoretical 6.7% to practical 2-3%.

Environmental Considerations

Deep ocean water is cold, nutrient-rich (high in nitrogen, phosphorus, and silica), and low in pathogens. Bringing it to the surface has multiple effects: nutrient upwelling can enhance marine productivity near the discharge, potentially supporting fisheries. But it also introduces cold water into warm surface environments, potentially disrupting coral reefs and tropical marine ecosystems. Large-scale OTEC deployment could even affect local ocean circulation patterns. Environmental impact assessment is essential for any OTEC project.

Economic Reality and Island Nations

OTEC electricity costs $0.15-0.25 per kWh — far more expensive than onshore solar ($0.03-0.05) or wind ($0.03-0.06) in most markets. But for island nations dependent on imported diesel at $0.30-0.50 per kWh, OTEC is already economically competitive. It also provides 24/7 baseload power (unlike solar and wind), produces desalinated water as a byproduct, and the cold water discharge can support air conditioning and cold-water agriculture. For tropical island nations, OTEC may be the most comprehensive clean energy solution available.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify nine key components of the OTEC system: surface water temperature, deep water temperature, temperature differential, fluid flow rate, turbine efficiency, energy conversion, environmental impact, infrastructure cost, and power output.

E **Establish Relationships**

Students map relationships showing that surface and deep water temperatures determine temperature differential, which constrains Carnot efficiency, which determines how much fluid flow is needed for target power output — with cost and environmental impact scaling with flow rate.

V **Visualize & Model**

Students build a computational model showing the nine-component OTEC system with thermodynamic efficiency as the central constraint.

E **Evaluate Outcomes**

Students run tropical optimal, reduced differential, and maximum scale scenarios to identify the conditions under which OTEC transitions from thermodynamically possible to economically viable.

R **Revise & Extend**

Students add desalinated water output, cold water agriculture, or biofouling rate to model more complete OTEC system economics.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"The ocean absorbs more energy from the Sun in one hour than humanity uses in a year. What if we could tap into it? We can — but thermodynamics has some humbling news for us."*

Watch For

Show the stunning scale of ocean thermal energy: enough to power civilization hundreds of times over. Then reveal the catch: Carnot efficiency limits extraction to a tiny fraction.

Slide 2: Learning Objectives

3 min

Say: *"Today you are engineering a power plant that runs on warm water and cold water — no fuel, no emissions, 24 hours a day, 365 days a year. But the physics is brutally unforgiving."*

Watch For

Have students read objectives. Pre-teach 'Carnot efficiency' and 'working fluid.' Demonstrate the concept: if you could only capture 3% of the heat from a campfire, how big a fire would you need to warm a room?

Slide 3: Big Question

3 min

Say: *"The energy is there — an almost infinite amount. The technology works — it was demonstrated in 1930. So why are we not using it? The answer is in the math of thermodynamics."*

Watch For

Quick-write: Students calculate Carnot efficiency for a 20 degree Celsius temperature difference. Most will be shocked at how low it is. Discuss what this means for practical energy extraction.

Slide 4: LEVER Framework

2 min

Say: *"We are building a model that bridges thermodynamics, ocean science, engineering, and economics. The answer to whether OTEC can power cities depends on all four."*

Watch For

Preview LEVER steps. Emphasize that this model tests whether an elegant physics concept can survive contact with engineering reality and economic constraints.

Slide 5: Activity 1: Components

8 min

Say: *"Nine variables. Two you set from ocean conditions. Seven that the thermodynamics and economics determine. The question is whether the numbers work."*

Watch For

Guide students through all nine components. Spend extra time on temperature differential and its relationship to Carnot efficiency — this sets the ceiling for everything else.

Slide 6: Activity 2: Connections

8 min

Say: *"Small temperature difference means low efficiency. Low efficiency means you need huge water flow. Huge water flow means enormous pipes and pumps. Enormous infrastructure means high cost. Follow the cascade."*

Watch For

Help students trace the cascade from thermodynamic constraint to economic consequence. Show how the 2-3% net efficiency drives every other number in the system.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let us see if the numbers work. Can a 20 degree Celsius temperature difference power a city? Run the scenarios and find out."*

Watch For

Students predict Power Output and cost for each scenario. Run simulations. The reduced differential scenario shows how rapidly viability drops with even small temperature changes.

Slide 8: Discoveries

5 min

Say: *"OTEC works. The physics is sound. But the Carnot limit is merciless — it forces every other number in the system to be enormous to compensate for the tiny efficiency."*

Watch For

Lead discussion comparing OTEC to other renewables. Where does OTEC make sense? (Tropical islands, baseload, co-products.) Where does it not? (Temperate regions, competing with cheap solar.)

Slide 9: STEM Challenge

12 min

Say: *"A Pacific island nation pays $0.40 per kWh for diesel electricity. Design an OTEC system that beats that price while providing clean water and cold-water agriculture. Can you make it work?"*

Watch For

Groups design comprehensive OTEC island energy systems. Include electricity, desalination, and cold water agriculture revenue. Present with economic analysis from model data.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Surface Water Temperature, Deep Water Temperature
- **INTERNAL:** Temperature Differential, Fluid Flow Rate, Turbine Efficiency, Energy Conversion, Environmental Impact, Infrastructure Cost, Power Output

Relationships

- **Temperature Differential to Energy Conversion:** POSITIVE (+) — Larger temperature differential increases the Carnot efficiency limit, allowing more thermal energy to be converted to useful work per unit of water processed. Each additional degree of differential improves theoretical efficiency by approximately 0.3 percentage points.
- **Fluid Flow Rate to Power Output:** POSITIVE (+) — Higher fluid flow rates process more thermal energy per second, directly increasing gross power output. However, pumping costs also increase with flow rate, so the relationship between flow rate and NET power output has diminishing returns.
- **Fluid Flow Rate to Environmental Impact:** POSITIVE (+) — Higher flow rates mean more cold, nutrient-rich deep water is brought to the surface and more warm surface water passes through the system. This increases both the beneficial effects (nutrient enrichment) and harmful effects (thermal disruption, ecosystem alteration) on the surrounding marine environment.

Key Discoveries

- The Carnot efficiency limit for OTEC is brutally low — about 6.7% for a 20 degree Celsius differential — meaning the system must process enormous volumes of water to extract useful energy, driving up infrastructure cost and environmental impact
- Net energy conversion (after pumping costs) is only 2-3% of thermal energy processed, requiring flow rates of millions of gallons per minute per 100 MW — making the pipes and pumps the dominant engineering challenge
- Environmental impact is a double-edged variable: nutrient-rich deep water brought to the surface can enhance marine productivity (positive) but also disrupts existing ecosystems and temperature patterns (negative)
- OTEC provides uniquely valuable baseload power — unlike solar and wind, the temperature differential exists 24 hours a day, 365 days a year, making it one of the most consistent renewable sources possible

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"The ocean has unlimited energy so OTEC can produce unlimited power"	While the ocean's thermal energy is enormous, the Carnot efficiency limit means only a tiny fraction can be extracted as useful work. A 20 degree Celsius temperature difference limits efficiency to about 7% theoretical and 2-3% practical. This means a 100 MW OTEC plant must process the same thermal energy as a 3,000+ MW fossil fuel plant — requiring massive infrastructure. Additionally, large-scale extraction would locally cool surface waters and warm deep waters, reducing the temperature differential and setting a natural upper limit on OTEC energy extraction in any given area.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"OTEC is just like any other heat engine — make it bigger and it works better"	OTEC is fundamentally different from conventional heat engines because its temperature differential is extremely small (20 degrees Celsius versus 500+ degrees Celsius for fossil fuel plants). This tiny differential means the system operates near the bottom of the Carnot curve where efficiency is extremely sensitive to temperature changes — a 5 degree Celsius reduction in differential can cut net power output by 50% or more. Scaling up helps reduce per-unit costs but cannot overcome the thermodynamic constraint. Size helps economics but not physics.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Pumping cold water from the deep ocean has no environmental impact"	Deep ocean water is fundamentally different from surface water: it is cold (4-5 degrees Celsius), dense, nutrient-rich (high nitrogen, phosphorus, silica), and supports different biological communities. Bringing millions of gallons per minute to the surface introduces cold, nutrient-rich water that can disrupt tropical surface ecosystems adapted to warm, nutrient-poor conditions. The nutrients can trigger algae blooms. The cold discharge can stress coral reefs. However, these impacts can also be beneficial — the nutrients support enhanced fisheries and the cold water enables aquaculture. The environmental impact is complex and site-specific, not negligible.